

Final Review

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- PS 6 & 7 should be getting returned today

Grading Breakdown

ENVS 423L	
Homeworks	50%
Midterm	25%
Final	25%

EPS 522	
Homeworks	30%
Midterm	20%
Final	20%
Term Project	30%

Exam Details

- Tuesday, December 9 from **7:30-9:30 am** in this room
- 14 questions, 2 hour exam time
- Several are lifted directly from problem sets
- Several are slight modifications from problem sets
- The rest are drawn from concepts from the lecture slides
- You will want a calculator!

Topics Covered

- Streamflow and Runoff Processes
- Vadose Zone
- Groundwater

Streamflow and Runoff Processes

- Stream discharge measurement
- Baseflow separation
- Runoff generation mechanisms
 - ▶ Three categories, and subcategories within them
 - ▶ Climate/geology/land use factors that control runoff generation type
- Stream network
- Concentration-discharge Relationships

Vadose Zone

- Material properties of soil
- Grain size distribution and soil textural class
- Water content and pressure
- Relation of suction and hydraulic conductivity with water content
- Infiltration and factors affecting it
- Redistribution

Groundwater

- Hydraulic head components and calculation
- Darcy's law
- Aquifer types and storage properties
- 3-D groundwater flow equation
 - ▶ What are the assumptions and how does the equation change by changing those assumptions?
- Regional groundwater flow
- Flownets, flowlines, lines of equipotential
 - ▶ How to interpret for aquifer thickness/slope/geology
 - ▶ How to interpret for SW/GW interactions
- Groundwater water balance components

"Cheat Sheet"

You are allowed to bring an 8.5x11" sheet of white printer paper with handwritten notes, equations, etc. on one side.

You will need to turn this in with the exam, so please write your name at the top of it.

Questions?

You got this!

